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Introduction to LLMs and Their Working

This report explores the foundational architecture of LLMs and how they work behind the screens of the computers of an everyday consumer. Covering topics ranging from tokenisation to various LLMs available today and their specific use cases.

Core Concepts:

LLM

A Large Language Model (LLM) is an advanced architecture trained on massive amounts of data to predict, recognise and generate human-like responses. LLMs work on Transformer architecture, relying generally on statistical probability to determine the most logical word in the sequence based on the given context. Due to their training on the mentioned vast amounts of various types of data, they can develop a generalised semantic understanding of languages, allowing for text summarisations, translations, complex code generation and logical reasoning.

The key power of a transformer is that it can process entire sentences at once, rather than analysing them slowly word by word. This allows the model to map how words relate to each other across vast distances in a text, giving it a grasp of context. The transformer relies on two critical components:

Embeddings: Conversion of raw words into vectors, where words with similar meanings are grouped.

Attention Mechanisms: Calculates a mathematical weight between every word in the text. Using attention weights, the transformer accurately predicts logical steps in a sequence.

The LLM's operational life is divided into 2 phases:

Training: The expensive, compute-heavy phase where the model is built from scratch, happening once in a massive server farm.

Inference: The phase where the model uses pre-existing education to run the user's prompt and generate a response.

API

An Application Programming Interface (API) is the intermediary that allows developers to connect private codebases directly to AI infrastructure hosted by large providers such as OpenAI, Anthropic, or Groq. The API allows a developer to send a lightweight POST request over the internet; the host handles the heavy computation and returns a JSON response, thereby democratising AI development.

Working of LLMs

LLMs operate purely on mathematics. When a user enters a prompt, it passes through a rigorous processing pipeline:

Tokenisation: The string is broken into text fragments called tokens; they may be a whole word or might just be a syllable. This is done using advanced algorithms like Byte-Pair Encoding (BPE) or WordPiece.

Vocabulary Matching: Every token is mapped to a unique integer ID in the model's vocabulary list.

Vector Embeddings: These IDs are projected onto a continuous mathematical space called the embedding layer. Here, a token is represented by a dense vector, allowing the model to capture semantic relationships.



Token Pricing

Tokens can carry fundamentally different economic valuations depending on the core deployment infrastructure.

Free Tokens: Companies release Open-Weight Models, such as Llama by Meta which run locally on the user's own machine. There is no metering or monthly payments because the computations are all performed on the user's own machine (GPU), and the cost is shifted instead onto the hardware costs and electricity required to run the machine.

Paid Tokens: Commercial API providers bill users on a scale per 1 million tokens processed. Input tokens are cheaper because the model reads them simultaneously; however, output tokens can be more expensive because they require sequential compute-heavy autoregressive loops; hence are billed.

Major LLM Companies

OpenAI

GPT-5.6 Sol: OpenAI's premier model, designed for logical reasoning and advanced multi-step computer automation capabilities. It's best used for agentic workflows, financial modelling, and cybersecurity vulnerability testing.

GPT-5.6 Terra: A mid-tier model that offers an optimal balance between advanced intelligence and economical infrastructure execution costs. Most useful for everyday code debugging, data extraction and deep document analysis.

OpenAI o3: Created for heavy reasoning that utilises reinforcement learning to “think” for long stretches before responding; used generatively for mathematical answers, scientific hypotheses generation, and hard technical problem solving.

Anthropic

Claude Fable 5: It's the absolute top-tier model, setting benchmarks in multi-step task handling and raw long-horizon logical autonomy. It's uniquely suited for scientific research, defensive cybersecurity mapping, and deploying autonomous AI software agents.

Claude Opus 4.8: Opus is the flagship model known for accurate instruction-following and algorithmic problem solving.

Claude Sonnet 4.6: Anthropic's highly responsive mid-tier model is designed for delivering a balance of rapid speed, lower operational costs and multimodal vision capabilities. It's ideal for enterprise applications, processing image data for automated UI testing.

Google

Gemini 3.5 Flash: Google's latest engine, built with ultra-fast execution speeds and a native 1-million token context window; best for autonomous multi-step execution loops, rapid high-volume text transformations and data processing.

Gemini 3.1 Pro: A high-end reasoning model designed to maintain precision and context tracking across dense data inputs and hence best used for corporate PDF data dumps, reviewing multiple codebases all at once.

How Groq is Different

Groq is essentially an AI Inference Provider; it does not design or train its own models, bypassing months of training a model on massive parallel computations.

Groq takes already completed flagship open-source models (Meta's LLaMA) and hosts them on its own platform to generate a response for a user.

Where traditional AI infrastructure relies on GPUs, Groq bypasses this by inventing a specialised chip called the Language Processing Unit – LPU. Thus, allowing Groq to run massive models at blistering speeds, often exceeding 500-1000 tokens per second.